

Module-4

1. Significance of MAC Layer and its Position in OSI Model

The **MAC (Medium Access Control)** layer is a sublayer of the **Data Link Layer (Layer 2)** in the OSI model.

It controls how devices access the wireless medium and transmit frames without collisions.

The MAC layer handles addressing, frame delivery, retransmissions, acknowledgments, roaming, authentication, and power saving.

In Wi-Fi, the MAC layer is very important because many devices share the same radio channel.

It works between the LLC layer above and the PHY layer below.

2. 802.11 MAC Header Frame Format and Field Purpose

The 802.11 MAC frame contains several fields:

Field	Purpose
Frame Control	Identifies frame type/subtype
Duration/ID	Reserves medium using NAV
Address 1	Receiver address
Address 2	Transmitter address
Address 3	Source/Destination/BSSID
Sequence Control	Tracks fragments and ordering
Address 4	Used in WDS/bridging
QoS Control	QoS priority information
HT Control	High throughput features
Frame Body	Actual payload
FCS	Error detection using CRC

The MAC header helps devices properly route and process wireless frames.

3. MAC Layer Functionalities in Management, Control, and Data Planes

Management Plane Functions

These establish and maintain Wi-Fi connections.

Functions:

- Beacon transmission
- Scanning
- Authentication
- Association/Reassociation
- Roaming
- Probe requests/responses
- Synchronization

Control Plane Functions

These help manage medium access.

Functions:

- RTS/CTS
- ACK/Block ACK
- NAV management
- Frame sequencing
- Retransmissions
- CSMA/CA operation

Data Plane Functions

These carry actual user traffic.

Functions:

- Data transmission
- Encryption/decryption
- QoS handling
- Frame aggregation
- Traffic forwarding

4. Scanning Process and Its Types

Scanning is how a client discovers available Wi-Fi networks.

Two types:

Passive Scanning

The client listens for Beacon frames sent periodically by APs.

Process:

1. AP sends beacon
2. Client listens
3. Client discovers SSID and capabilities

Advantages:

- Less traffic
- Lower power usage

Disadvantage:

- Slower discovery

Active Scanning

The client actively sends Probe Request frames.

Process:

1. Client sends probe request
2. AP responds with probe response
3. Client selects AP

Advantages:

- Faster discovery

Disadvantage:

- More traffic and power consumption

5. Client Association Process

The Wi-Fi connection process happens step-by-step.

Step 1: Scanning

Client discovers APs.

Step 2: Authentication

Client and AP exchange authentication frames.

Step 3: Association

Client requests to join AP.

Step 4: Security Setup

802.1X or WPA handshake occurs.

Step 5: IP Address

DHCP assigns IP address.

After this, normal communication begins.

6. EAPOL 4-Way Handshake and Derived Keys

The 4-way handshake is used in WPA2/WPA3 after authentication.

Purpose:

- Generate encryption keys
- Verify both client and AP know the PMK

Step 1

AP sends:

- ANonce

Step 2

Client generates:

- SNonce
- PTK

Then sends MIC to AP.

Step 3

AP verifies MIC and sends:

- GTK
- Confirmation

Step 4

Client confirms installation of keys.

Handshake complete.

Keys Used

Key	Purpose
PMK	Master key
PTK	Unicast encryption
GTK	Broadcast/multicast traffic
MIC	Integrity protection

7. Power Saving Scheme in MAC Layer

Wi-Fi devices save battery using MAC layer power-saving mechanisms.

The client enters sleep mode when inactive.

AP buffers packets for sleeping clients.

The client periodically wakes up to check for buffered data.

Power Saving Mechanisms

Legacy Power Save

Uses TIM/DTIM beacons.

U-APSD

Used in VoIP devices for efficient battery saving.

TWT (Target Wake Time)

Wi-Fi 6 feature allowing scheduled wake times.

This greatly improves battery life in IoT devices.

8. Medium Access Control Methodologies

Wi-Fi uses **CSMA/CA**.

Steps:

1. Device listens to channel
2. If busy → wait
3. Random backoff starts
4. Device transmits
5. Receiver sends ACK

Collision avoidance is used because collision detection is difficult in wireless networks.

Additional methods:

- RTS/CTS
- NAV
- Backoff timers

9. Block ACK Mechanism and Advantages

Instead of acknowledging every frame separately, Block ACK acknowledges multiple frames together.

Process:

1. Sender transmits multiple frames
2. Receiver sends one Block ACK bitmap

Advantages:

- Reduces overhead
- Improves throughput
- Better efficiency for high-speed Wi-Fi

Used heavily in:

- 802.11n/ac/ax

10. A-MSDU, A-MPDU, and A-MSDU in A-MPDU

Frame aggregation improves efficiency by combining multiple packets.

A-MSDU (Aggregate MAC Service Data Unit)

Multiple MSDUs combined into one MPDU.

Advantages:

- Less header overhead

Disadvantage:

- Entire aggregate retransmitted if error occurs

A-MPDU (Aggregate MAC Protocol Data Unit)

Multiple MPDUs grouped together.

Advantages:

- Individual retransmissions possible
- Better reliability

A-MSDU inside A-MPDU

Combination of both techniques.

Structure:

- Multiple MSDUs inside MPDU
- Multiple MPDUs aggregated together

This provides:

- Very high efficiency
- Reduced overhead
- Better throughput in modern Wi-Fi networks